Zheng Fang

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SUMMARY

Computer Science graduate with experience building AI-powered systems, scalable backend services, and full-stack web applications. Proficient in Python, Java, Go, and C++, with hands-on experience developing RAG-based AI systems, computer vision pipelines, and distributed cloud applications. Seeking a New Graduate Software Engineer (SDE) full-time role starting in 2026.

EDUCATION

Harvard University **Cambridge, MA**
M.Des (Technology track) Cross-registered at MIT EECS *Sep. 2024 - Jun. 2026* Coursework: Fundamentals of Programming, Data Science I & II, ML Operations, Linear Algebra, Probability and Statistics, Data Visualization.

Syracuse University **Syracuse, NY**
Bachelor of Architecture - Magna cum laude *Sep. 2017 - Jun. 2022*

TECHNICAL SKILLS

Programming Languages: Java, JavaScript/TypeScript, Python, Golang, SQL, C, C++, R
Databases and Cloud: MySQL, PostgreSQL (Cloud SQL), Amazon RDS, AWS Lambda/EC2, Google GCE/GAE, Docker, Kubernetes (GKE), Vertex AI
Web Development: Spring, Java Servlet, AngularJS, Node.js, jQuery, HTML & CSS, React JS, Redux, Linux
AI / ML & Data: LLM Integration (OpenAI API), Retrieval-Augmented Generation (RAG), Recommendation Systems, Computer Vision (YOLO, CLIP), Vector Search (FAISS), Data Modeling & Feature Engineering

PROJECT EXPERIENCES

TopShot: An AI Photography Tutor with a RAG-Based Vision + Language Pipeline
- Developed an AI photography tutor that analyzes user-uploaded photos and delivers technical and composition feedback, powered by a **RAG-backed vision + language pipeline** over a curated dataset of ~25K photos.
- Built the backend with **FastAPI** and **PostgreSQL (Cloud SQL)**, covering EXIF metadata extraction, image embedding, and exemplar retrieval to ground feedback generation on **Vertex AI**.
- Implemented the **TypeScript** frontend for photo upload and interactive feedback display.
- Containerized backend and frontend with **Docker Compose** and deployed to **Google Kubernetes Engine (GKE)** via **Artifact Registry**.
- Set up **CI/CD with GitHub Actions**, automating linting (ruff), pytest unit/integration/e2e tests, image builds, and GKE deployment on every push to main.

OnlineOrder: A Spring Boot Based Online Food Ordering System
- Developed CRUD REST APIs with **Spring Controllers**, including registration, menu search, ordering, and checkout.
- Utilized **Spring Data JDBC** for **PostgreSQL** database access on **AWS RDS** (menus, restaurants).
- Implemented **Spring Security** for session-based authentication.
- Employed a structured architecture (controller, service, repository) with dependency injection.
- Built frontend using **ReactJS and Ant Design** for cart management and orders.
- **Containerized and deployed on AWS App Runner** for scalability.

Agent AI: Full-Stack RAG & MCP-Based Document Q&A Agent
- Designed and developed a full-stack AI agent using **React and Ant Design**, integrated with voice features via **react-speech-recognition** and **speak-tts** for seamless spoken interaction.
- Implemented a **Retrieval-Augmented Generation (RAG) pipeline** leveraging **GPT-5** and **LangChain** components to enable semantic document comprehension and context-aware responses.
- Designed a hybrid search strategy combining RAG outputs with web search through a **Model Context Protocol (MCP) server**, standardizing tool access and integrating results from **SerpAPI (Google Search API)**.
- Built and deployed a backend service with **Node.js and Express**, supporting document ingestion, query handling, and API orchestration.

SocialAI: An AI-Based Social Network
- Designed and implemented a social network web application with **ReactJS**.
- Used **OpenAI's DALL·E 3** to assist users to create and update posts.
- Improved the authentication using token-based registration/login/logout flow with **React Router v4** and server-side user authentication with **JWT**.
- Launched a scalable web service in **Go** to handle posts and deployed to **Google Cloud (Google App Engine)**.
- Used **Elasticsearch** (deployed to **GCE**) to provide search functions such that users can search recent posts and list personal posts.

RESEARCH AND PUBLICATIONS

- Fang, Z., Li, X.
  Prompts to Perception: A Promptable Segmentation Approach to Transparency and Safety.
  CAADRIA 2026, Full Paper Accepted, 2025.
- Fang, Z., Tan, Y., Hoberman, C.
  TopoFold: Transformable Structure Generation from Topologically Complex Surfaces via Normal Intersection Search.
  CAADRIA 2026, Full Paper Accepted, 2025.
- Anonymous Authors
  Vinema: A Tangible, Vinyl-Inspired Device for Human–AI Musical Co-Creation.
  Manuscript Under Review, 2025.
- Fang, Z., Li, X., Sun, M., Tian, R.
  Sectional Sampling Controlled Generation of High-Resolution Architecture Design with Diffusion Model and Latent Interpolation.
  ACADIA 2024, 2023.